Patient-Specific Finite Element Analysis of Stent Deployment for Sinus Venosus Atrial Septal Defects

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Introduction

The management of sinus venosus atrial septal defects (SVASDs) remains a significant challenge in congenital cardiology. Although covered stent correction (CSC) offers a less invasive treatment alternative, its clinical success can be compromised by complications such as stent migration, pulmonary vein (PV) compression or persistent shunt flow [1]. A precise understanding of device–tissue interactions is therefore essential for effective intervention planning. This study aimed to develop a patient-specific finite element (FE) model to quantify how stent position, sizing, and the use of a second anchoring stent influence biomechanical outcomes, and thereby identify the most favourable deployment strategy for the patient.

Methods

A patient-specific SVASD geometry was reconstructed from ECG-gated CT imaging and discretised using a multi-resolution shell-element mesh. The venous walls were represented as linear elastic ($E = 0.4$ MPa, $\nu = 0.48$), while a 10-zig Cheatham-Platinum (CP) stent was modelled as elasto-plastic and meshed with beam elements. Simulations were run in a commercial FE solver (Abaqus/Explicit). The in situ deployment protocol comprised crimping, balloon-driven expansion, balloon deflation, caudal flaring, and final recoil. A matrix of twelve simulations was defined by combining three longitudinal stent positions along the superior vena cava (SVC; proximal, baseline and distal), two sizing strategies (nominal diameter and 10% radial oversizing), and two stent configurations (single-stent and dual-stent). For each simulation, three outcomes were extracted: stent anchorage within the SVC, estimated from nodal contact pressure (CPRESS); PV compression, derived from the logarithmic minimum principal strain; and defect coverage, evaluated qualitatively through direct visual assessment.

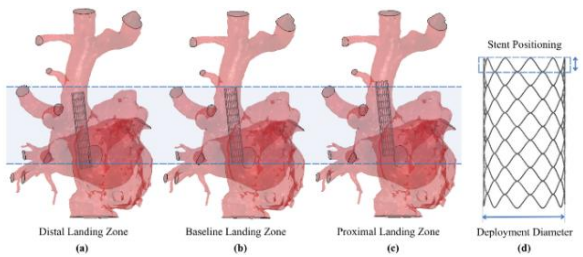


Figure 1: (a–c) Stent deployment positions within the patient-specific anatomy. (d) Schematic representation of the CP stent.

Results

Evaluation of the twelve simulations showed that transcatheter SVASD correction success is highly dependent on device positioning within the anatomy. Proximal placement in a dual-stent configuration yielded the highest CPRESS values, reflecting the strongest mechanical fixation within the SVC and consistent with its larger, more uniform landing zone. In single-stent configurations, radial oversizing reduced the contact area at both baseline and distal sites, suggesting that overexpansion in non-uniform regions may promote malapposition. Proximal deployment was also the only strategy to consistently achieve complete defect coverage, whereas baseline and distal positioning more often resulted in partial coverage due to stent foreshortening. None of the configurations exceeded the 12% strain threshold established as a conservative safety margin, indicating a minimal risk of significant venous obstruction in this patient.

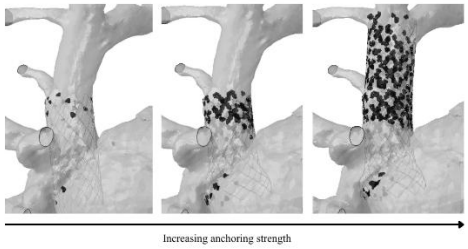


Figure 2: Progression of stent–vessel contact area.

Conclusions

This patient-specific in silico framework highlights FEA as a non-invasive tool for pre-procedural planning in CSC of SVASDs, extending beyond conventional imaging to anticipate device–tissue mechanical behaviour. Future work should adopt more representative vascular material models [2], incorporate fluid-structure interactions [3], and undergo validation across a larger patient cohort to confirm its clinical reliability.

References

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